

Downstream Mach Number M_2

For a normal shock : $M_1 > 1 \rightarrow M_2 < 1$

Formula
(perfect gas) : $M_2^2 = \frac{1 + \frac{\gamma-1}{2} M_1^2}{\gamma M_1^2 - \frac{\gamma-1}{2}}$

"The stronger the incoming, supersonic flow, the harder the shock knocks it down"

Standard
Values :

$M_1 = 1.2$	$\rightarrow$	$M_2 = .84$
$M_1 = 2.0$	$\rightarrow$	$M_2 = .58$
$M_1 = 3.0$	$\rightarrow$	$M_2 = .48$

Solid for intuition.

Static Pressure Ratio

Formula : $\frac{P_2}{P_1} = 1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1)$

" Stronger shock
= Stronger Pressure
Jump "

$$\frac{P_2}{P_1} = 4.5$$

$$P_2 = 4.5 P_1$$